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Permian and Triassic palaeolatitudes of the Ukrainian shield with implications for Pangea reconstructions

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SUMMARY

A collection of 306 oriented samples from 43 Permian and Triassic intrusions from the Ukrainian Shield, the southwest portion of the East European Craton, has yielded valuable palaeomagnetic directions with new ages obtained using the $^{40}Ar/^{39}Ar$ method. Andesitic intrusions have Late Triassic ages (six dating samples) of 204.2 ± 1.6 Ma to 215.7 ± 2.0 Ma and dual-polarity Dec/Inc = 60.1°/+64.4°, $k = 96$, $\alpha_{95} = 4.5°$, derived from $N = 12$ sites. Trachyte dykes have an early Artinskian (mid-Early Permian) age (one dating sample) of 282.6 ± 2.6 Ma, and yielded Dec/Inc = 204.3°/−23.8°, $k = 27.7$, $\alpha_{95} = 6.5°$, from $N = 19$ sites. The palaeolatitude of the trachyte intrusions is 12.4° ± 3.7° N.

Because the Artinskian and younger Permian palaeopoles obtained from the Gondwana continents are subject to uncertainties, some studies have adopted a palaeopole (at 41°S, 61°E) largely based on results from ∼280 Ma Permian quartzporphyries in the Southern Alps of Italy as proxy to position Gondwana at that time. With this palaeoposition and our new ∼280 Ma palaeolatitude for Baltica, a Pangea A reconstruction cannot be supported. To avoid its inherent continental overlap of Africa and Eurasia, a Pangea B reconstruction has been favoured, wherein Eurasia's southern margin faces the northcoast of South America instead. However, the 280 ± 10 Ma palaeomagnetic data from the Gondwana continents themselves give a mean palaeopole at 30°S, 59°E, which results in a palaeogeographic position of Gondwana that allows a Pangea A type reconstruction. Thus, the choice between Pangea A versus B in the Artinskian hinges on data selection and reliability criteria and the assumptions about tectonic coherence of northern Adria (i.e. the Southern Alps) with Africa in Permian times. A more reliable mid-Early Permian palaeopole from cratonic Gondwana would provide a more definitive conclusion.

Key words: Palaeomagnetism applied to tectonics.

1 INTRODUCTION

It has long been known that Permian and Early Triassic palaeomagnetic data are not in good agreement between Gondwana and Laurussia and imply an impossible overlap of some 11° of latitude when the continents are longitudinally constrained as in conventional Pangea reconstructions (De Boer 1965; Irving 1967, 1977; Muttoni *et al.* 2003, Fig. 6; Van der Voo & Torsvik 2004, Fig. 7). To solve this conundrum an alternate reconstruction called Pangea B (Fig. 1) has been proposed that places Gondwana and Laurussia in non-competing areas (Irving 1977; Bachtadse *et al.* 2002; Muttoni *et al.* 2009). Pangea A, the configuration proposed by Alfred Wegener (Weneger 1912), refined by Bullard *et al*. (1965), and often cited in textbooks, juxtaposes the eastern US coast next to NW Africa. In Pangea B, the Eastern United States is next to NW South America instead (Irving 2004). The Atlantic Ocean started opening in the late Early Jurassic (Schettino & Turco 2009) and all agree that by that time Western Europe was longitudinally connected to the north coast of Africa in a Pangea A type fit. To transform Pangea B into Pangea A before the Jurassic, a dextral mega-shear of some 3500 km is necessary between Gondwana and Laurussia (Fig. 1). However, little geological evidence exists for this mega-shear (Hallam 1983; McElhinny & McFadden 2000, pp. 300–301; Torsvik & Cocks 2004), regardless of whether its movements occurred primarily in the Triassic (Irving 1977; Morel & Irving 1981; Torcq *et al.* 1997) or in the Permian (Muttoni *et al.* 1996, 2003, 2009).

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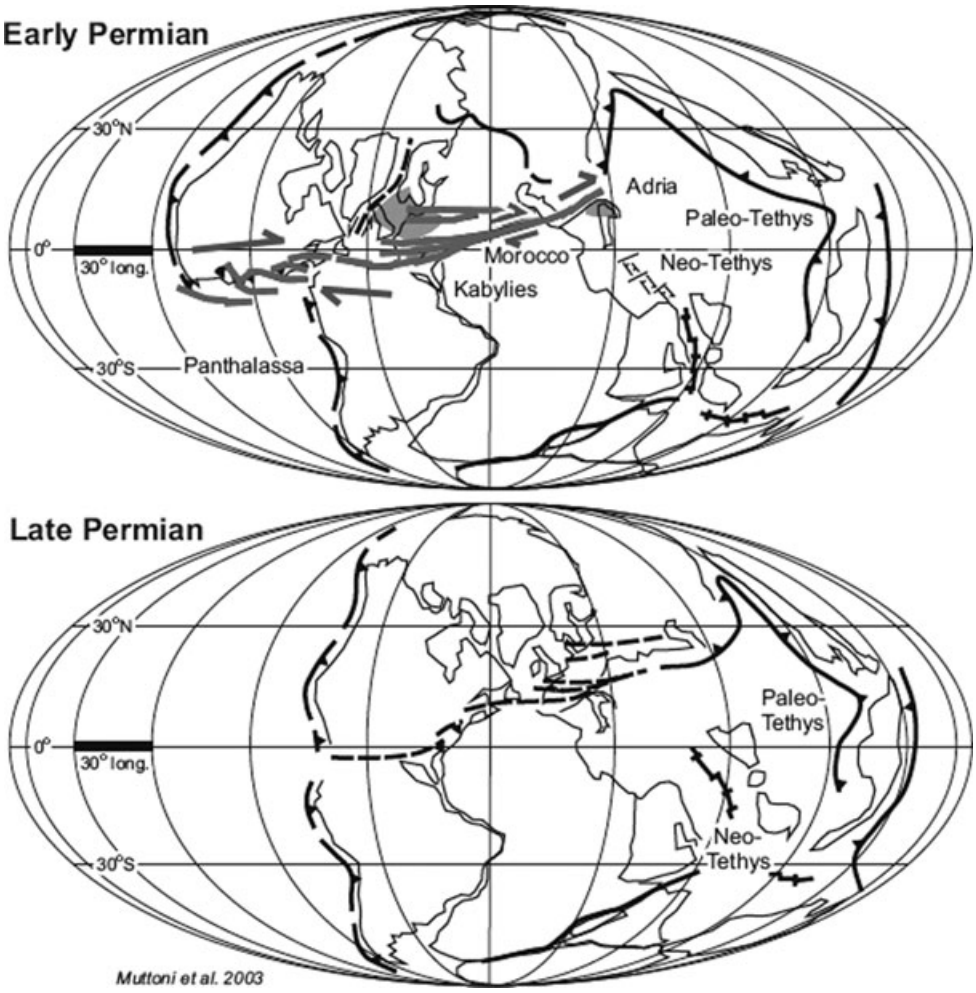


Figure 1. Pangea B (top panel), changing into Pangea A (bottom) along a WSW–ENE dextral mega-shear zone of some 3500 km length between Laurussia and Gondwana (from Muttoni *et al.* 2003, reprinted with permission from Elsevier).

The eleven-degree overlap may be caused by magnetizations that suffered inclination shallowing in sedimentary rocks. Therefore, collecting igneous rocks could help solve this problem (Rochette & Vandamme 2001). Non-dipole fields and other geomagnetic field complexities have also been proffered as possible causes (Van der Voo & Torsvik 2001; Vizán & Zele 2007). This paper applies Occam's razor, that is, identifying the simplest solution as usually being correct, in the expectation that higher quality palaeomagnetic results collected from critical areas will substantiate the original scheme of Pangea A's reconstruction (Van der Voo & Torsvik 2004) and will render the Pangea B model unnecessary.

The structural stability of the Ukrainian shield makes an ideal sampling area because it has remained tectonically undisturbed since the Permian, as explained in the geology and sampling section. As a result, no tilt correction, which could complicate the study, needs to be applied to the palaeomagnetic results.

2 GEOLOGY AND SAMPLING

A total of 306 Permian and Triassic samples were collected from 43 intrusions and a couple of sites in their baked-contact rocks from the eastern terminus of the Ukrainian Shield. This area (Fig. 2a) forms part of the Palaeozoic Baltica continent that evolved from the Meso-Proterozoic East European Craton (EEC; Mikhailova & Kravchenko 1986), which formed 1.8–1.7 billion years ago when Fennoscandia, Sarmatia and Volga-Uralia amalgamated (Fig. 2a inset). Extension and rifting, starting as early as 1.65 Ga and continuing into the Palaeozoic, affected the belt between the Ukrainian Shield and Voronezh Massif, in a basin called the Dniepr-Donets Graben and the Donbas Foldbelt to its east (Fig. 2a). The EEC became incorporated as Baltica into Rodinia 1.2–0.9 billion years ago (Bogdanova *et al.* 2008) and after the breakup of Rodinia followed by independent drift during the earlier Palaeozoic, it became part of Laurussia in the Silurian and then Pangea in the Carboniferous.

The Dniepr-Donets-Donbas rifting did not succeed in fragmenting Sarmatia, and hence can be considered as a classic example of an aulacogen (= failed rift; Shatsky 1946). However the rift caused the Voronezh Massif in the north and the Ukrainian Shield in the south, joined together since the Achaean (Shchipansky & Bogdanova 1996), to become two separate shield areas (Glasmacher *et al.* 2004). During Hercynian and Uralian mountain building, there were compressional events that caused the earlier formed basins to invert, but these did not greatly impact the shield areas. Weak rifting in the basins followed during the Permian and Triassic, which resulted in andesitic and trachytic magmatism (Nikishin *et al.* 1996). The Ukrainian Precambrian shield areas are suitable for palaeomagnetic studies of Permian and younger rocks, because they have remained tectonically stable since the Carboniferous.

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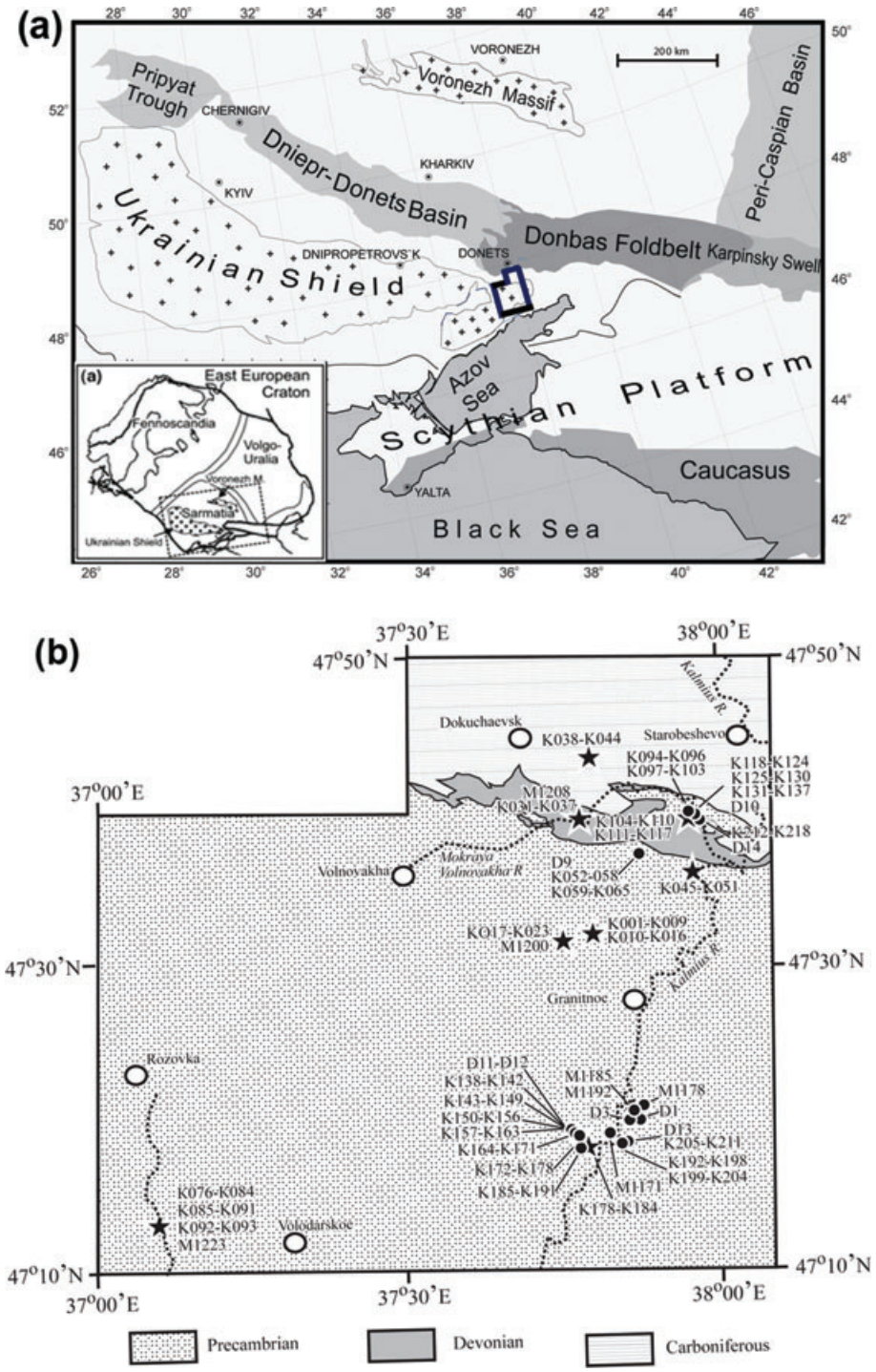


Figure 2. (a). Schematic geological map showing the outcrop areas of the Voronezh Massif and the Ukrainian shield, as well as the intervening Dniepr-Donets Graben and the Donbas fold belt (adapted from Stephenson *et al.* 2006, with permission of Elsevier). Inset: map of the divisions of the East European Craton (Danišík *et al*. 2008) and location of the Ukrainian Shield. The sampling area, in the easternmost end of the Ukrainian Shield between the Azov Sea and the Donets Basin, is outlined, and is shown in (b) with sampling site locations (filled stars for andesite sites and filled circles for trachyte sites).

A total of 43 trachytic and andesitic sites were drilled in the field (Fig. 2b), where each site is assumed to yield a separate recording of the ancient magnetic field; two baked-contact sites were also sampled. The collection was divided into three batches, which were measured by the three different groups of researchers from the United States, Russia and Ukraine, respectively. A solar compass was used to spot check the magnetic direction at some sites, but the compass was damaged in the field. Fortunately, the magnetic and available solar directions agree with each other, because apparently the magnetization of the samples is not so large as to affect the orientation with a compass. A correction, taken from the International Geomagnetic Reference Field (National Oceanic and Atmospheric Administration, website, 2010), has been applied to the field measurements to offset the present-day local magnetic declination of 7°E.

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3 PALAEOMAGNETIC AND AGE DATING METHODOLOGIES

In the laboratory the cylindrical field-drilled palaeomagnetic samples were sawed to specimens with 2.2 cm height. The samples studied at the University of Michigan were stored and treated inside the magnetically shielded room of its palaeomagnetic laboratory, to minimize the acquisition of a present-day viscous magnetization. The natural remanent magnetization (NRM) of the samples was measured using a three-axis 2G magnetometer. The specimens were thermally demagnetized in an ASC TD-48 oven in the shielded room where the residual field did not exceed 200 nT.

In the palaeomagnetic laboratory of the Geological Institute of the Russian Academy of Science in Moscow, cubic specimens of 8 cm^3 volume were sawed from hand blocks. One specimen from each hand-sample was stepwise demagnetized in 15–20 increments up to 700°C in a homemade oven with internal residual fields of approximately 10 nT and measured with a JR-4 spinner magnetometer with a noise level of 0.05 mA m^{-1}; the magnetometer was placed into large Helmholtz coils to reduce the ambient field.

Measurements were processed with a Labview module called MichCryo7 that plots the data as vector endpoints in orthogonal demagnetization diagrams (Zijderveld 1967a) and as intensity versus applied temperature, as well as in stereonets. The directions of linear vectorial trajectories were computed using the computer program Super-IAPD (Torsvik 2000), which uses principal component analysis (Kirschvink 1980). In cases where endpoints were not stable but plotted along great circle paths, the software program Palaeomac (Cogné 2003) was used. This program combines stable-endpoint observations and intersections of great-circle trajectories (McFadden & McElhinny 1988). Site means were calculated with Super IAPD and Fisher (1953) statistics, and palaeomagnetic pole positions were calculated with Palaeomac. Site coordinates, site-mean directions and the associated statistical parameters, are listed in Table 1; overall mean directions and palaeopoles are given in Table 2.

$^{40}Ar/^{39}Ar$ age dating and mineral separation used standard techniques in the $^{40}Ar/^{39}Ar$ Geochronology Laboratory of the Norwegian Geological Survey (NGU) in Trondheim. Before packing in Al-foil, mineral separates were hand-picked under a binocular microscope and all samples were rinsed in alternating acetone and distilled water. The sample packets were stacked and loaded in a sealed Al-capsule with Cd-shielding for irradiation in the 5C site at the McMaster Nuclear Reactor facility, Hamilton, Canada. The samples were irradiated at McMaster for 16 hr 40 min at 3 MW (50 MWH) with nominal neutron flux of 4×10^{13} n cm^{-2} s^{-1}; nominal temperature in the irradiation site is <50 °C (M. Butler, personal communication 2001). Production of isotopes from Ca and K were determined by irradiation of CaF and K2SO4 salts; values of 36/37Ca = 0.000169, 39/37Ca = 0.000736 and 40/39K = 0.032593 were used. Neutron fluence (the neutron flux integrated over the duration of the irradiation) was monitored with Tinto biotite of 410.3 Ma (Rex & Guise 1995). We incorporated a conservative 1 per cent error in *J*-value for all unknowns.

Gas from irradiated samples was released in a stepwise fashion from a resistance furnace. Furnace conditions are similar to those described in Eide *et al.* (2002). Gas released from a sample at a single temperature step was cleaned in the extraction line for 11 min using two pairs of SAES AP-10 getters, mounted in isolated sections of the line, each maintained with their own vacuum pumps. The purified gas was then analysed on a MAP 215–50 mass spectrometer. Data for blanks, monitors and unknowns were collected on a Johnson electron multiplier with gain setting at 1, while the magnet was automatically scanned over masses 35 through 41 in a cycled, 'peak-hop' mode. Masses 37–40 were each measured in 10 cycles and 10 counts per mass per cycle; mass 36 was measured with 20 counts per cycle.

Dynamic blank measurements on mass 40 indicate a stable background (1.0×10^{-13} ccSTP). Background levels (blanks) for the furnace were measured at 100–200 °C temperature increments prior to each sample analysis. Furnace blanks were maintained at levels of $<1.1 \times 10^{-11}$ ccSTP for mass 40 and $3–5 \times 10^{-14}$ ccSTP for mass 36 at temperatures of 500–1000 °C; blanks increased to 3.0×10^{-11} ccSTP for mass 40 and 1.1×10^{-13} ccSTP for mass 36 at high temperatures (1200–1400 °C). Background levels of masses 37 and 39 did not change significantly from dynamic blank levels at any temperature (1×10^{-13} ccSTP for mass 37; $<5.3 \times 10^{-14}$ ccSTP for mass 39). Background levels for mass 38 were $<3 \times 10^{-14}$ ccSTP at all temperatures. At experimental temperatures between 600 and 1000 °C, furnace blanks for mass 40 typically were <1 per cent of the sample signal size.

Data from unknowns were corrected for blanks prior to being reduced with the IAAA (Interactive Ar-Ar Analysis) software package (Visual Basic programming for PC Windows) written by T.H. Torsvik and N.O. Arnaud and based in part on equations in Dalrymple *et al.* (1981) and McDougall & Harrison (1999). Data reduction in IAAA incorporates corrections for interfering isotopes, mass discrimination (measured with an air pipette), error in blanks and decay of ^{37}Ar.

4 AGE DATING RESULTS

Eight andesite and one trachyte sample (M1171) yielded hornblende crystals that could be irradiated for age dating. Of the eight andesite samples (M1199–M1234), six gave meaningful results, as summarized in Table 3. The dating places the intrusion of the andesites in the Late Triassic between 204.2 ± 1.6 Ma and 215.7 ± 2.0 Ma. One characteristic andesite age spectrum (sample M1234), which corresponds to site K111-K117, is shown in Fig. 3 and its isochron is shown in Fig. 4.

The four steps at about 210 Ma in Fig. 3 represent 79 per cent of the argon released and they form an accurate and unambiguous plateau. The other four andesite samples with plateau ages all have similarly good behaviour; the ages all fall within 6 Myr of the mean of 210 Ma (Table 3); one additional sample yielded a very compatible weighted-mean age of 204.2 ± 1.6 Ma (Table 3), and thus helps to confirm that the andesites are Late Triassic in age.

In contrast, the trachytes contained silicates that were visibly altered, so that we could obtain reasonably fresh hornblende from only one sample (from site M1171) for age dating purposes. Its age spectrum is shown in Fig. 5. Its weighted-mean age of 282.6 ± 2.6 Ma indicates that the trachytes are mid-Early Permian (Artinskian) in age. Six steps in the argon release spectrum are chronologically very close to each other and differ by less than 10 Myr. The age is a weighted mean based on 82.7 per cent of accumulated ^{39}Ar released. We will see below that the andesites and trachytes each show well-grouped directions of magnetization that are quite distinctly different from each other. We feel justified, therefore, to assign all andesite sites to the Late Triassic and all trachytes to the mid-Early Permian.

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Table 1. Site-mean magnetization directions and statistical parameters.

Site	*n*	*Z*	GPS lat (N)	GPS long (E)	*D*°	*I*°	*k*	α_{95}°
Andesites and one baked contact (site K104-K110)								
K001-K009	9	9	47°32.144′N	37°47.925′E	226.5	−61.3	186.4	3.8
K010-K016	7	6	47°32.154′N	37°47.915′E	221.1	−58.6	104.9	6.6
K017-K023	7	7	47°31.749′N	37°44.960′E	57.1	69.2	111.4	5.7
K031-K037	7	7	47°39.730′N	37°46.548′E	60.5	60.3	172.3	4.6
K038-K044	6	6	47°43.659′N	37°47.571′E	55.6	51.7	717.6	2.5
K045-K051[c]	7	4	47°36.234′N	37°57.461′E	*172.8*	*76.6*	*339.5*	*5.0*
K076-K084	8	8	47°13.179′N	37°05.966′E	267.8	−63.1	136.4	4.8
K085-K091	6	5	47°13.215′N	37°05.957′E	253.2	−64.0	519.9	3.6
K092-K093[a]	2	2	47°13.215′N	37°05.985′E	*235.1*	*−59.8*	*82.5*	*27.8*
K104-K110	6	6	47°39.627′N	37°57.358′E	224.6	−67.1	240.4	3.9
K111-K117	7	7	47°39.627′N	37°57.358′E	236.7	−71.7	93.4	7.0
K178-K184[b]	7	0	47°18.251′N	37°47.097′E	*No stable remanence*			
M1200	7	6	47° 31.737′N	37° 44.968′E	56.5	65.9	92.5	7.0
M1208	7	6	47° 39.763′N	37° 46.407′E	63.3	66.8	127.0	6.0
M1223	7	7	47° 13.265′N	37° 05.852′E	262.1	−64.5	118.2	5.6
Site	*n*	*Z*	GPS lat (N)	GPS long (E)	*D*°	*I*°	*k*	α_{95}
Trachytes and one baked contact (site K094-K096)								
K052-K058[a]	4	3	47°37.408′N	37°52.321′E	*324.4*	*70.0*	*11.3*	*38.5*
K059-K065	6	4	47°37.388′N	37°52.252′E	199.3	−36.0	56.1	12.4
K094-K096[a]	3	3	47°39.655′N	37°57.265′E	*193.0*	*−26.8*	*10.2*	*40.8*
K097-K103	7	7	47°39.655′N	37°57.265′E	213.2	−38.6	144.2	5.0
K118-K124[a]	6	6	47°39.732′N	37°58.105′E	*288.4*	*3.6*	*5.4*	*28.7*
K125-K130[a]	4	3	47°39.732′N	37°58.105′E	*22.8*	*68.7*	*12.1*	*37.1*
K131-K137[a]	7	3	47°39.690′N	37°57.948′E	*212.4*	*−32.7*	*5.5*	*58.8*
K138-K142	5	4	47°19.169′N	37°45.434′E	191.4	−12.7	57.6	12.2
K143-K149[c]	7	4	47°19.218′N	37°45.610′E	*204.7*	*3.1*	*205.8*	*6.4*
K150-K156	7	7	47°19.216′N	37°45.680′E	220.3	−16.5	31.3	11.1
K157-K163	7	5	47°19.176′N	37°45.711′E	215.0	−4.4	18.3	18.4
K164-K171	8	6	47°19.031′N	37°45.915′E	214.0	−13.2	25.9	13.4
K172-K177	6	5	47°18.828′N	37°46.401′E	208.8	−11.6	46.4	11.3
K185-K191[a]	7	4	47°18.225′N	37°46.984′E	*213.9*	*5.9*	*12.7*	*26.9*
K192-K198	7	6	47°18.490′N	37°50.582′E	196.7	−21.1	140.3	5.7
K199-K204	5	5	47°18.597′N	37°50.555′E	194.9	−17.4	93.4	8.0
K205-K211	6	6	47°18.691′N	37°50.698′E	188.6	−16.2	159.5	5.3
K212-K218[b]	7	0	47°39.315′N	37°58.353′E	*No stable remanence*			
M1171	7	6	47° 18.992′N	37° 49.132′E	217.6	−40.7	109.4	6.4
M1178	7	6	47° 20.848′N	37° 52.367′E	197.6	−14.4	87.3	7.2
M1185	7	6	47° 20.531′N	37° 51.677′E	217.0	−26.5	140.9	5.7
M1192	7	7	47° 20.55′ N	37° 51.748′E	206.8	−37.1	101.7	6.0
D1-HT	10	7	47° 20.00′ N	37° 52.0′ E	205.8	−24.0	22.2	13.1
D3[a]	7	4	47° 20.00′ N	37° 51.0′ E	*195.4*	*−18.0*	*15.8*	*23.8*
D9[a]	7	5	47° 37.383′N	37° 52.250′E	*201.9*	*−37.5*	*10.8*	*24.4*
D10–5d1	7	6	47° 39.683′N	37° 57.950′E	180.4	−26.2	14.9	18.2
D13	6	5	47° 18.680′N	37° 50.645′E	193.6	−26.2	540.7	3.3
D14-HTC	6	6	47° 39.303′N	37° 58.297′E	203.5	−43.1	205.1	4.7
D11&D12	5	4	47° 19.210′N	37° 45.713′E	218.6	−18.4	275.0	5.6

D°, declination angle; *I*°, inclination angle; *k*, precision parameter; α_{95} (°), radius of 95 per cent confidence cone around mean direction, *n*, number of samples demagnetized; *Z*, number of samples used for calculation/analysis. A declination correction of 7.2° has been applied to all results.

[a]Sites with *k* lower than 10, α_{95} higher than 20°, or with directions from only one or two samples, have been excluded and entered in italics.

[b]Sites where the magnetization was eliminated for >95 per cent at <200°C, entered in italics.

[c]Sites where sample directions still appear to be contaminated by overprints, entered in italics.

Andesite site K024-K030 had mislabelled samples and was therefore not studied or measured.

5 ANDESITES: PALAEOMAGNETIC RESULTS

The andesite samples were collected from blocky, solid and relatively unaltered intrusions, thus rendering them promising for palaeomagnetism. Thermal demagnetization showed that most of the samples had very small or no overprint magnetizations that may have been introduced by recent growth of a viscous remanence. Typical examples of thermal demagnetization of andesite samples with nearly univectorial decay are shown in Fig. 6, which reveal the characteristic southwesterly and up, or northeasterly and down direction of this suite of intrusions.

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Table 2. Mean directions and palaeopoles.

Lithology	D°	I°	k	$\alpha_{95}{}^\circ$	N	Pole Lat (°)	Pole Long (°)	Pole dp	Pole dm
Andesites (reversed)	241.4	−65.4	75.1	7.0	7				
Andesites (normal)	58.4	62.8	128.7	6.8	5				
Andesites (combined)	240.1	−64.4	96.0	4.5	12	50.0 N	106.4 E	5.8	7.2
Trachytes	204.3	−23.8	27.7	6.5	19	49.4 N	179.7 E	3.7	6.9

Overall mean directions of the andesite and trachyte collections and palaeopoles with the semimajor and semiminor axes of the oval of 95 per cent confidence (dm, dp). Other explanations as in Table 1.

Table 3. Results of the $^{40}Ar/^{39}Ar$ age dating of hornblendes.

Sample	J-value	Steps used (Temp)	^{39}Ar (per cent)	Spectrum (Ma)	1σ (excl.)	1σ) (incl. J)	Inv. isochron age (Ma)	1σ (excl. J)	1σ (incl.)	MSWD	$^{40}Ar/^{36}Ar$	1σ
M1171	0.005881	6 to 7 987–1055 °C	82.7	**282.6** Weighted mean	**0.05**	**2.6**						
M1199	0.005880	6 to 11 986–1117 °C	89.7	**204.2** Weighted mean	**0.03**	**1.6**						
M1208	0.005834	8 to 11 1024–1102 °C	83.8	209.0 Plateau age	0.05	2.0	**209.2**	**1.0**	**2.2**	1.36	293.6	5.7
M1213	0.005878	10 to 13 1085–1263 °C	94.8	208.2 Plateau age	0.08	1.9	**209.3**	**0.1**	**2.0**	0.02	283.3	7.4
M1215	0.005860	9 to 12 1032–1153 °C	89.7	209.4 Plateau age	0.06	2.0	**210.7**	**1.2**	**2.3**	2.43	285.4	10.5
M1226	0.005884	10 to 13 1079–1211	58.3	214.9 Plateau age	0.02	2.0	**215.7**	**0.4**	**2.0**	0.37	278.6	17
M1234	0.005857	7 to 10 920–1073 °C	79.2	209.2 Plateau age	0.05	2.0	**208.9**	**0.8**	**2.1**	1.99	296.5	3.6

Results of the $^{40}Ar/^{39}Ar$ age dating of hornblendes from one trachyte sample (M1171) and six andesite samples. Preferred ages in bold. For M1171, the weighted mean age is based on two large, consecutive steps. For M1199 we report a weighted mean age because the spectrum displays more variation than can be explained by the analytical uncertainty alone. Two other samples, M1207 and M1216, did not yield geologically interpretable results.

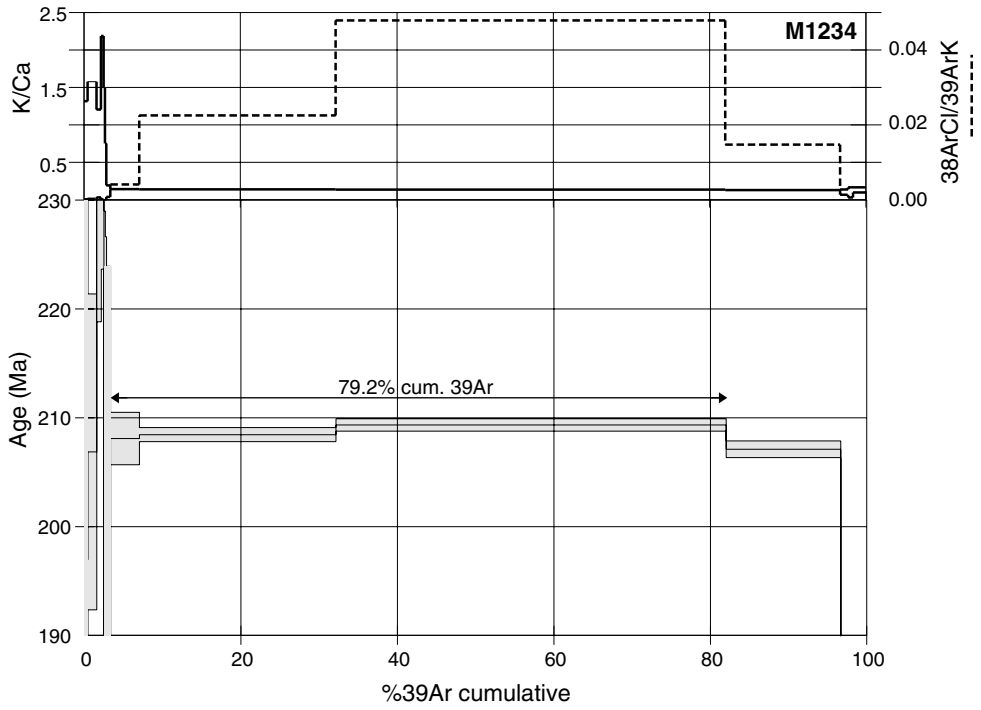


Figure 3. Age spectrum (2σ) of an andesite dyke sample. The plateau age of 209.2 ± 2.0 Ma (1σ, including J uncertainty) comes from steps 7–10 at 920–1073 °C where 79.2 per cent of the argon is released. Top panel: K/Ca and 38ArCl/39ArK versus cumulative 39Ar.

On close inspection, however, it can be seen from the remanence-versus-temperature plots (Figs 6b, c and e) that there are actually two sets of magnetic mineral grains that carry the characteristic directions. One set unblocks above 440 °C and below about 580 °C at which point about 60 per cent of the remanence is lost, and the other unblocks in the interval 580–675 °C. For sample K113B, a very small overprint can be seen from 0 to 250 °C (Fig. 6a), which is unlikely to be an ancient component. The

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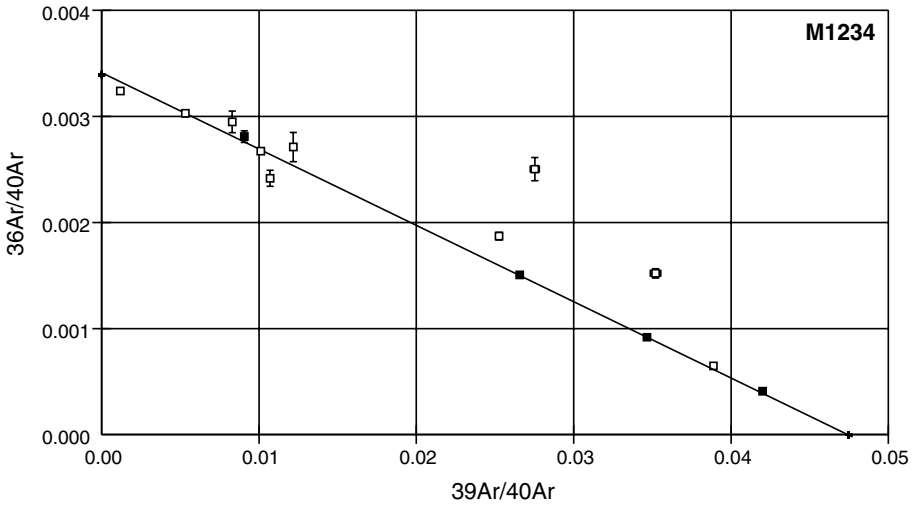


Figure 4. An inverse isochron age of 208.9 ± 2.1 Ma (1σ, including error on J value) for the same andesite sample as shown in Fig. 3. There is a 40/36 intercept of 296.5 ± 3.6 (overlapping atmospheric level at 95 per cent confidence level). Black squares correspond to the steps included in the plateau in Fig. 3 (79.2 per cent of cumulative 39Ar).

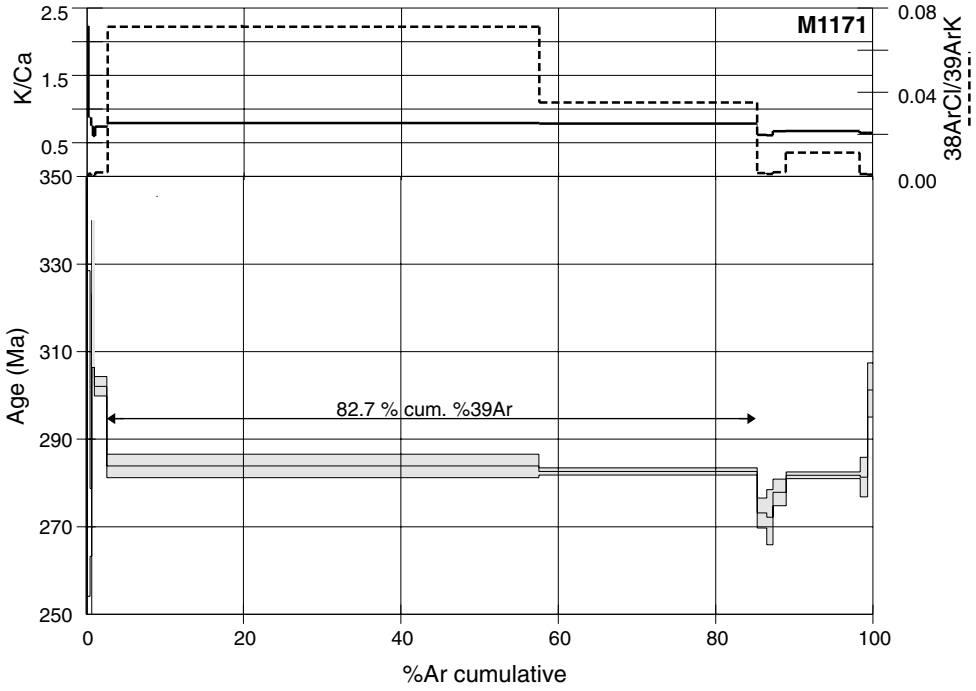


Figure 5. Age spectrum (2σ) of a trachyte dyke sample from site M1171, showing a weighted-mean age of 282.6 ± 2.6 Ma (1σ, including J uncertainty), based on steps 6 and 7 (987—1055 °C), which represent 82.7 per cent of accumulated ^{39}Argon released (Table 3). Top panel: K/Ca and 38ArCl/39ArK versus cumulative 39Ar.

intermediate-temperature component (ITC) decays between 420 and 580 °C, whereas the high-temperature component (HTC) is removed in the interval 580–675 °C. The ITC's carrier is likely magnetite, whereas the HTC's carrier must be haematite. The ITC and HTC components in this sample are both southwesterly and up; they have similar and ancient (Late Triassic) directions of magnetization.

For sample K014B (Figs 6c and d), no low-temperature component is revealed, the decay is nearly univectorial. Its remanence also consists of two components, one being removed with sharp decay between 500 and 560 °C and the other between 580 and 670 °C. Most of the andesite sites show behaviour in a similar fashion.

Other intrusions have nearly antipodal NE and steeply down magnetizations (Figs 6f and 7). In sample K021A (Fig. 6f), the decay is nearly univectorial. The remanence intensity versus temperature plot (Fig. 6e) shows that the remanence has unblocking temperatures that are again characteristic of a magnetization carried by both low-Ti magnetite and by haematite, with the latter eliminated well above 585 °C. Both components have northeasterly and down directions in this normal-polarity sample.

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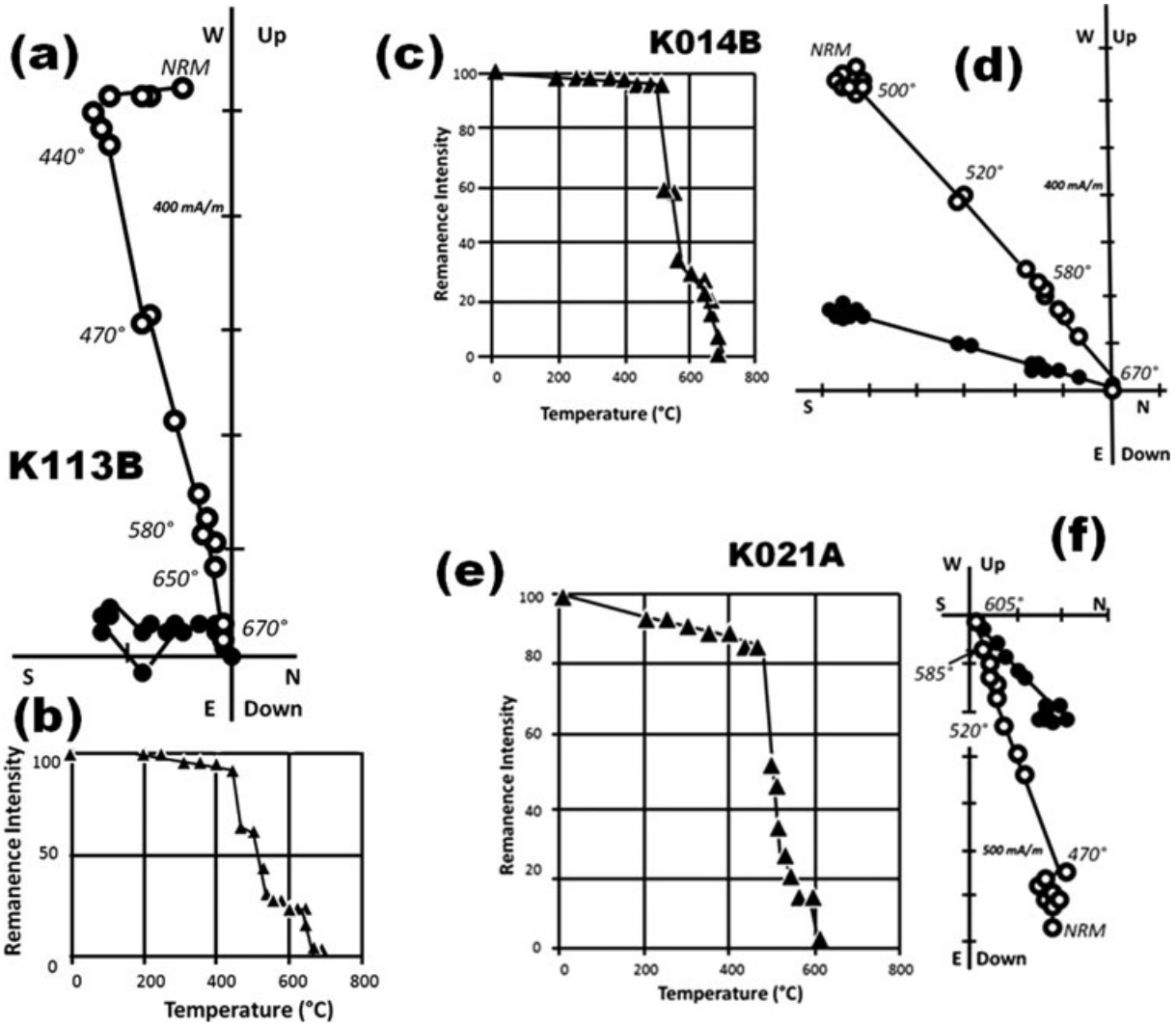


Figure 6. Plots of normalized intensity (to 100 per cent), as measured during thermal demagnetization of andesite samples, are presented in (b, c and e). Orthogonal projections of the endpoint of the magnetization vector (Zijderveld 1967a), as measured after each thermal treatment, are presented in a, d and f. Open (full) symbols represent projections onto the vertical (horizontal) plane. (a, b) Thermal demagnetization of andesite sample K113B. (c, d) Thermal demagnetization of andesite sample K014B. (e, f) Andesite sample K021A, which has the opposite polarity from that of samples K014B and K113B. NRM intensity values are 528 mA m^{-1} for K113B, 877 mA m^{-1} for K014 and 745 mA m^{-1} for sample K021A.

The unblocking temperatures indicate that Ti-poor magnetite and haematite are the most likely carriers of these two components. Site-mean directions of the SW/up and NE/down directions are shown in Fig. 7 and are logically interpreted as representing reversed and normal polarity magnetizations. Where magnetite and haematite both carry the characteristic remanence, not only are their directions the same, but also their polarity; thus, we have no reason to think that the magnetite and haematite magnetizations are of different age. The normal and reversed mean directions, derived from the andesite sites ($N = 12$), pass Tauxe's bootstrap reversal test (Tauxe 2010, p. 238), whereas the reversal test of McFadden & McElhinny's (1990) is also positive with classification B.

Three of the intrusions have either aberrant mean directions, too few samples per site or no measurable remanence above treatments at ∼200 °C (see footnotes *a*, *b* and *c* in Table 1).

The andesite site K111–117 intruded into Devonian basalts, where samples K104–110 were heated by the intrusion. The baked basalt samples have a site-mean direction (triangle in Fig. 7) that is statistically identical to those of the intruding andesite. In contrast, basalt samples farther away as well as a trachyte dyke located 5 m away from the andesite dyke have a different direction of magnetization; these are represented in Table 1 as sites K097–103 and K094–096. The magnetizations of these sites will be briefly discussed in the next section. Suffice for now to mention our conclusion that the contact of site K111–117 (baking samples K104–110), plus the recognition that trachytic host rocks farther away from the andesite dyke have not been reset, constitute a fully positive contact test.

As mentioned, $^{40}Ar/^{39}Ar$ dating of five samples places the andesites at ∼210 Ma (from 204.2 ± 1.6 Ma to 215.7 ± 2.0 Ma), representing the Late Triassic magnetic field with an overall mean andesite direction of Dec/Inc = 60.1°/+64.4°, $k = 96$, $\alpha_{95} = 4.5°$ (Fig. 7). The mean is based on 12 out of 15 sites. The means of the normal and reversed polarity site-mean directions are 177.1° apart.

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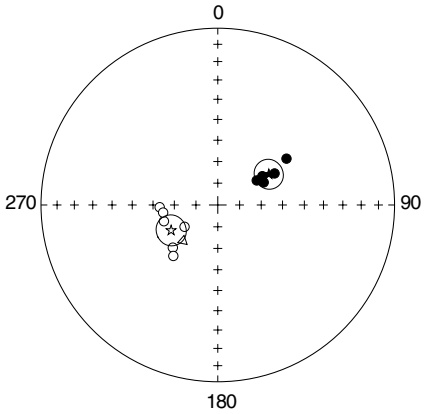


Figure 7. Equal-area projection of the andesite site means, showing normal ($N = 5$) and reversed-polarity ($N = 7$) directions. See Tables 1 and 2 for directions and statistical parameters. The means of the two nearly antipodal groups (stars with their cones of 95 per cent confidence) are 177.1° apart. The triangle represents the baked-contact site mean.

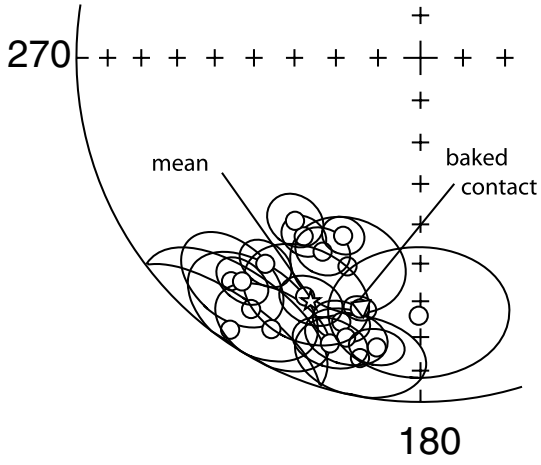


Figure 9. Equal-area projection of the trachyte site means (see Table 1). Ovals represent the cones of 95 per cent confidence around the means. The star is the mean direction of all the site-means, including the baked-contact site (inverted triangle at the end of the thin line).

The palaeolatitude calculated from this dual-polarity direction is 46.2°N and its palaeopole is located at 50°N, 106.4°E, with dp = 5.8 and dm = 7.2 (Table 2).

6 TRACHYTES: PALAEOMAGNETIC RESULTS

The aphanitic trachytes were more friable than the andesites but the trachytes were still suitable for palaeomagnetic drilling. Although the samples came from large blocky rock outcrops, the trachytes were generally more weathered (confirmed later in the demagnetization characteristics by relatively large overprints). The trachytes' different mineralogy and older age explain the larger overprints and weathering.

The trachytes have characteristic directions that are shallow to intermediate up and south-southwest (100 per cent reversed polarity) as can be seen in Figs 8(b) and (d). The site with samples K97-K103 is representative of the trachyte characteristics (Fig. 8). As was the case for the andesites, the remanence contributors are both haematite and magnetite (Figs 8a and c), which, importantly, both reveal the same direction. The mean direction (Table 2; Fig. 9) of 19 (out of 28) trachyte sites is Dec/Inc = 204.3°/–23.8°, k = 27.7, $\alpha_{95} = 6.5°$, $N = 19$. Excluded from the mean are sites with poor statistical parameters, lack of characteristic remanence above ~200 °C, or highly aberrant directions, as listed in italics in Table 1 and explained in its footnotes.

The trachyte site K097–K103 is intruded into a Devonian basalt and baked samples are K094–096. This is the same formation, and plausibly the same layer, as that sampled about 5 m away at site K104-K110 for the earlier mentioned contact test of an andesite dyke (for a schematic map of these sites, showing their spatial relationships, see online Appendix Fig. A1). The directions of the three samples K094, 095 and 096 of this Devonian basalt outcrop are similar to (but considerably more scattered than) those of the intruding trachyte (see Table 1) and appear to reflect a resetting by the Permian intrusion. This situation constitutes a contact test, but it can only be considered partial-positive, given that the nearby Triassic andesites are younger and do not qualify as 'host rock' to the trachyte intrusions.

Thus the mid-Early Permian geomagnetic field (at 282.6 ± 2.6 Ma) is interpreted to have a direction of Dec/Inc = 204.3°/–23.8°, k = 27.7, $\alpha_{95} = 6.5°$, $N = 19$, which indicates a palaeolatitude of 12.4° for the Ukrainian sampling area and a palaeopole at 49.4° N, 179.7° E, dp = 3.7, dm = 6.9.

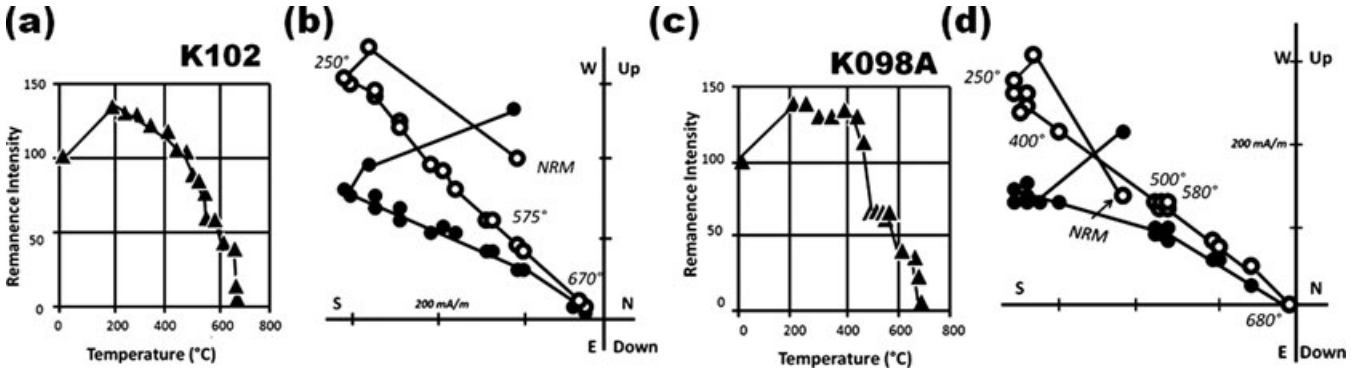


Figure 8. Plots of normalized intensity (to 100 per cent), as measured during thermal demagnetization of trachyte samples, are presented in (a) and (c). Orthogonal projections of the endpoint of the magnetization vector (Zijderveld 1967a), as measured after each thermal treatment, are presented in (b) and (d). Open (full) symbols represent projections onto the vertical (horizontal) plane. (a, b) Thermal demagnetization of trachyte sample K102. (c, d) Thermal demagnetization of trachyte sample K098A. NRM intensity values are 343 mA m^{-1} for K102 and 339 mA m^{-1} for sample K098A.

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7 DISCUSSION

The site-mean directions of the andesite and trachyte sites have been combined into overall mean directions (Table 2), from which palaeopoles have been computed. Our new poles can be compared to those of the global apparent polar wander path (APWP) in Baltica coordinates (Fig. 10a; Torsvik *et al.* 2008).

We argue that these new palaeopoles are reliable representations of the geomagnetic field at the time during initial cooling following intrusion. The sampling area is thought to have been tectonically stable since at least the Early Permian, there is no inclination error (igneous rocks cannot be affected by inclination shallowing) and no significant anisotropy of the dyke material was detected in 10 AMS measurements on pristine samples; moreover, there is a positive reversal test and a positive contact test for the Late Triassic andesite intrusions and a partial-positive contact test for the mid-Early Permian trachytes. The demagnetization trajectories gave no indication of contamination by partially overlapping overprint components in the sites accepted for calculation of the overall mean poles, whereas a few sites (e.g. K143–149) had to be rejected because of dominating overprints. The magnetic carriers are low-Ti magnetite and haematite. The possibility of late, post-intrusion oxidation is negated by the observation that the directions of the magnetizations carried by these two sets of mineral grains are invariably the same within measurement error.

Although the cones of 95 per cent confidence around the palaeopoles of this study overlap with the reference poles of the APWP for Baltica (Fig. 10a), the fact that our new palaeopoles fall visibly well off the main path could be taken to suggest some tectonic movement of the Ukrainian Shield with respect to other parts of Baltica. Recent studies, following the early work of Diego-Orozco & Henry (1993), have discussed evidence for rotations detected by Permian declination deviations within Baltica, that is, the part of Europe north of the Alpine belts (Bazhenov *et al.* 2008; Nawrocki *et al.* 2008; Taylor *et al.* 2009; Bazhenov & Shatsillo 2010). We note that the offsets of the andesite and trachyte palaeopoles are in opposite senses. These considerations caution us to postpone making conclusions about relative rotations until the database is more abundant and compelling.

The palaeolatitudes indicated by the trachyte and andesite results are not affected, of course, by rotations about vertical axes. The palaeolatitudes show that Ukraine moved from a low latitude (12.4°N) in the Early Permian to a much more northerly latitude by the Late Triassic (46.3°N) in a time span of about 70 Myr. This translates into a northward drift of slightly more than 5 cm yr^{-1}. Permian fossils verify that Ukraine was in a warm, equatorial climate at that time (White 1904; Ziegler 1990).

The palaeolatitude derived from the andesite, 46.3°N, is valid for the Norian stage (203.6–216.5 Ma) of the Late Triassic. The result is a valuable addition to the database, because not many results are available from igneous rocks of Norian age in Baltica. The palaeolatitude agrees with the predicted local latitude in Ukraine ($\lambda = 43.6°$N, $A_{95} = 4.7°$) from a commonly accepted Pangea-A1 palaeogeographic position at this time (Muttoni *et al.* 1996, their fig. 9), preceding the opening of the Atlantic at about 175 Ma by only 35 Myr. This result constrains any possible intra-Pangean relative movements (e.g. transformation of Pangea B or Pangea A2–Pangea A1) to the Carnian stage of the Late Triassic or earlier, as also concluded by Muttoni *et al.* (1996) and Irving (2004).

The palaeopole derived from the trachytes does not change the Permian palaeoposition of Baltica greatly. There are coeval palaeopoles of high quality (also derived from volcanics with good demagnetization and age control) that can be combined with our new result, and a range of combinations is documented in Table 4. The seven individual pole entries of Table 4 are plotted in Fig. 10(b).

All results with high-quality ages (those with AQ = 3 in Table 4, i.e. based on U/Pb or Ar-Ar methodologies on volcanics) are included insofar as their listed age, plus or minus its uncertainty, falls into the 290–270 Ma bin. We note that three results (poles 5–7) have older age spans that overlap only at 2σ. We therefore list mean

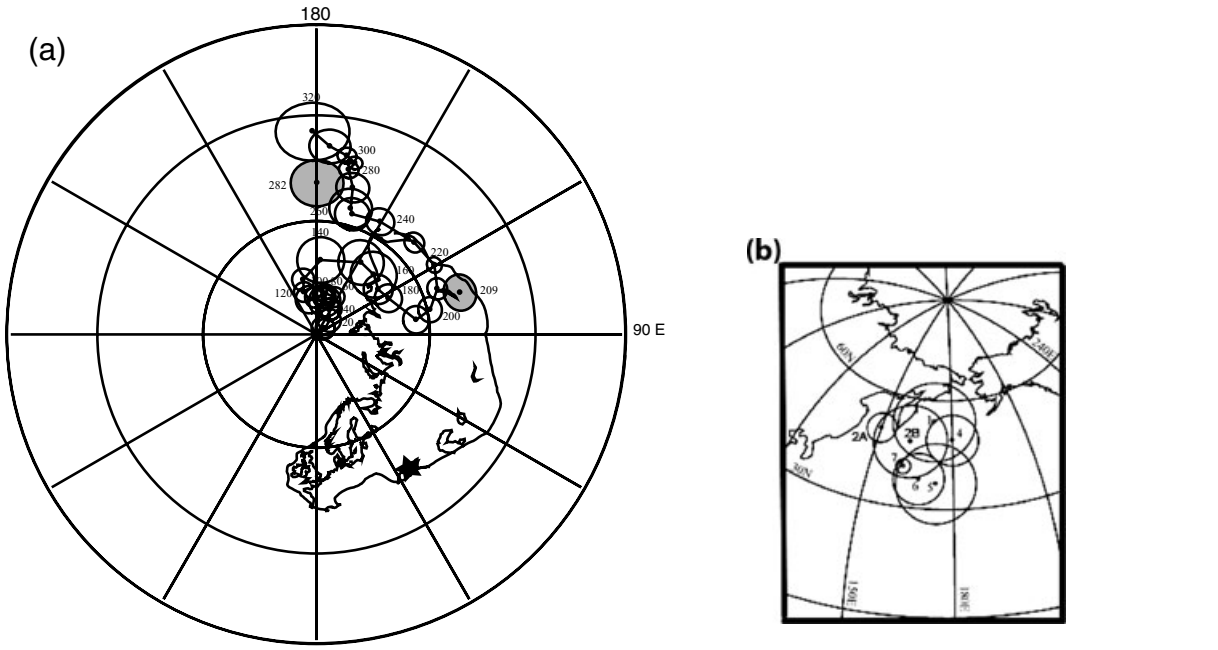


Figure 10. (a) The palaeopoles of this study (with shaded infilling of their cones of 95 per cent confidence) are plotted next to the global APWP in Baltica coordinates from Torsvik *et al.* (2008). The star represents the sampling locality. (b) Plot of our new mid-Early Permian palaeopole (#4) together with the other (approximately coeval) palaeopoles from Baltica, with the numbers corresponding to those of Table 4.

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Table 4. Mean palaeopoles for Baltica at 280 ± 10 Ma.

#	Formation	T/DC/AQ	Palaeopole	α_{95}	Age	Ref./palaeolat (λ)
1	Scania dykes	V/3/3	54N, 172E	11	279 ± 10	Bylund (1974)
2A	Exeter-1	V/4/3	50N, 150E	4	286 ± 7	Zijderveld (1967b)
2B	Exeter-2	V/3/3	48N, 163E	10	286 ± 7	Cornwell (1967)
3	Exeter average	V/3/3	49N, 157E		286 ± 7	(2A + 2B)
4	Ukraine trachytes	V/4/3	49N, 180E	7	283 ± 3	This study
5	Scania dolerites-1	V/4/3	37N, 174E	11	294 ± 8	Mulder (1971)
6	Scania dolerites-2	V/4/3	38N, 168E	7	294 ± 8	Bylund (1974)
7	Peterhead Dyke	V/4/3	41N, 162E	1	297 ± 12	Torsvik (1985)
[a] $N = 18$, Muttoni *et al.* mean pole		V	42.0N, 166.0E	3	273–294	λ = 10.5
$N = 3$, Mean of 1, 3 & 4		V/3/3	51.2N, 169.3E	12.3	280 ± 10	λ = 17.0
$N = 4$, Mean of 1, 2A, 2B & 4		V/3/3	50.9N, 166.1E	9.7	280 ± 10	λ = 17.9
$N = 7$, Mean of 1, 2A, 2B, 4, 5, 6, 7		V/4/3	45.7N, 167.1E	7.0	280 ± 10	λ = 13.1

T/DC/AQ means T = rock type (V = volcanics), DC = demag code and AQ = age quality from Van der Voo & Torsvik (2004).
Ref, reference and λ is the palaeolatitude calculated for Ukraine, at 47.4N, 37.8E.
[a]Muttoni *et al.* (2003) used this palaeopole for their Pangea analysis.

palaeopoles for Baltica that either include or exclude these three poles to examine their impact on the palaeolatitude of Baltica (λ in Table 4).

The palaeolatitudes calculated from the mean poles for Baltica, as extrapolated to our Ukrainian sampling area, range from 13.1°N to 17.9°N (Table 4). The table also includes the mean pole for Baltica that was used by Muttoni *et al.* (2003) in their arguments for Pangea B, and which would result in a palaeolatitude of 10.5°N for our Ukrainian sampling area. A recent study of Early Permian (Asselian, ∼293 Ma) sedimentary strata in the Donbas Foldbelt determined a palaeolatitude of 15.8°N (from two sites, each with the same palaeolatitude and $\alpha_{95} = 2.2°$; Meijers *et al.* 2010).

Rather than seek the blame for the mismatch of previously existing palaeomagnetic results with Pangea A type reconstructions entirely in the Baltica palaeopoles, improvements in the precision and reliability of mid-Early and Late Permian results from Gondwana are possible, necessary and potentially significant. The two Early Permian (poorly dated at ∼280 Ma) palaeopoles from Morocco (used by Muttoni *et al.* 2003) or the mean of 280 ± 10 Ma Gondwana palaeopoles listed by Torsvik & Van der Voo (2002), allow a tight Pangea A fit (Fig. 11). Laurentia's position, as derived from a 268–273 Ma palaeomagnetic result from igneous rocks in Illinois (Reynolds *et al.* 1997), also agrees reasonably well with Pangea A (see Fig. 11). New studies of the Late Permian and Early Triassic palaeomagnetic history of Gondwana are in progress on rocks from Argentina (see Domeier *et al.* 2009) and may further resolve the controversies surrounding Pangea reconstructions. Our analysis of the palaeomagnetic directions of the trachyte dykes shows that a large overprint in the trachytes can cause a downward shift in the inclinations, if thermal demagnetization did not succeed in removing it. If similar factors contributed to other palaeomagnetic results (notably for Gondwana), then the need for Pangea B type reconstructions will be an artefact.

Finally, it is interesting to examine the cause of the significant continental overlap of Europe and Gondwana in the reconstruction of Muttoni and colleagues (2003, 2009), given that their use of Baltica's mean pole for ∼280 Ma and the two results from Morocco's Early Permian volcanics is not different from the analysis of this paper. The main difference, however, is in the assumption that the mean palaeopole of the Early Permian rocks from the Southern Alps, part of the Mesozoic Adria promontory in Italy, is a good proxy for Gondwana's palaeopole. The palaeolatitude of the Southern Alps (∼8° N), as extrapolated from their combined Gondwana palaeopole (41N, 241E), fits comfortably within our reconstruction as shown by the dashed line in Fig. 11. However, if Africa and the rest of Gondwana are rigidly attached to this Adria microcontinent, as favoured by Muttoni *et al.* (1996, 2003), then the northernmost part of Gondwana will overlap significantly with southern Europe. The coherence of Adria and Gondwana has been argued by some for Jurassic and younger time, but is rather controversial for Permian times. Clearly, better evidence is needed about the Permian palaeogeography of Adria, as well as about the Permian APWP segment for Gondwana.

8 CONCLUSIONS

New palaeomagnetic results show that our sampling area in Ukraine was located at 12.4°N to 17.9°N at about 280 Ma, with the range depending on which results are included in the averaging. A result from Early Permian sediments in Ukraine has yielded a 15.8°N palaeolatitude (Meijers *et al.* 2010). With Gondwana in the palaeoposition used by Muttoni *et al.* (2003) for about 280 Ma, a palaeolatitude of 12°N of Baltica does not leave enough room for Pangea A, and therefore does not disprove Pangea B. However, with the somewhat higher palaeolatitudes from Table 4 (i.e. ∼17°N) for Baltica, and using the Gondwana palaeopole at 30°S, 54.9°E from Torsvik & Van der Voo (2002), a Pangea A type reconstruction becomes fully possible, because the latter Gondwana pole allows significantly more room for a Pangea A type fit, as shown in Fig. 11. The same conclusion is reached for the position of Gondwana based on the only results from igneous rocks in Africa (Morocco) with ages of 270–290 Ma. However, the latter results are lacking well-documented ages, and one has a surprisingly large scatter ($\alpha_{95} >$ 20°) in the results derived from the Mechra-Chougrane volcanics, Morocco (Westphal *et al.* 1979).

Although contributions from an octupole field cannot be ruled out (Van der Voo & Torsvik 2004), we now prefer to favour the idea that more reliable palaeomagnetic directions can solve the Pangea problem in the near future. Clearly, more reliable mid-Early to Late Permian palaeopoles for cratonic Gondwana will be desirable.

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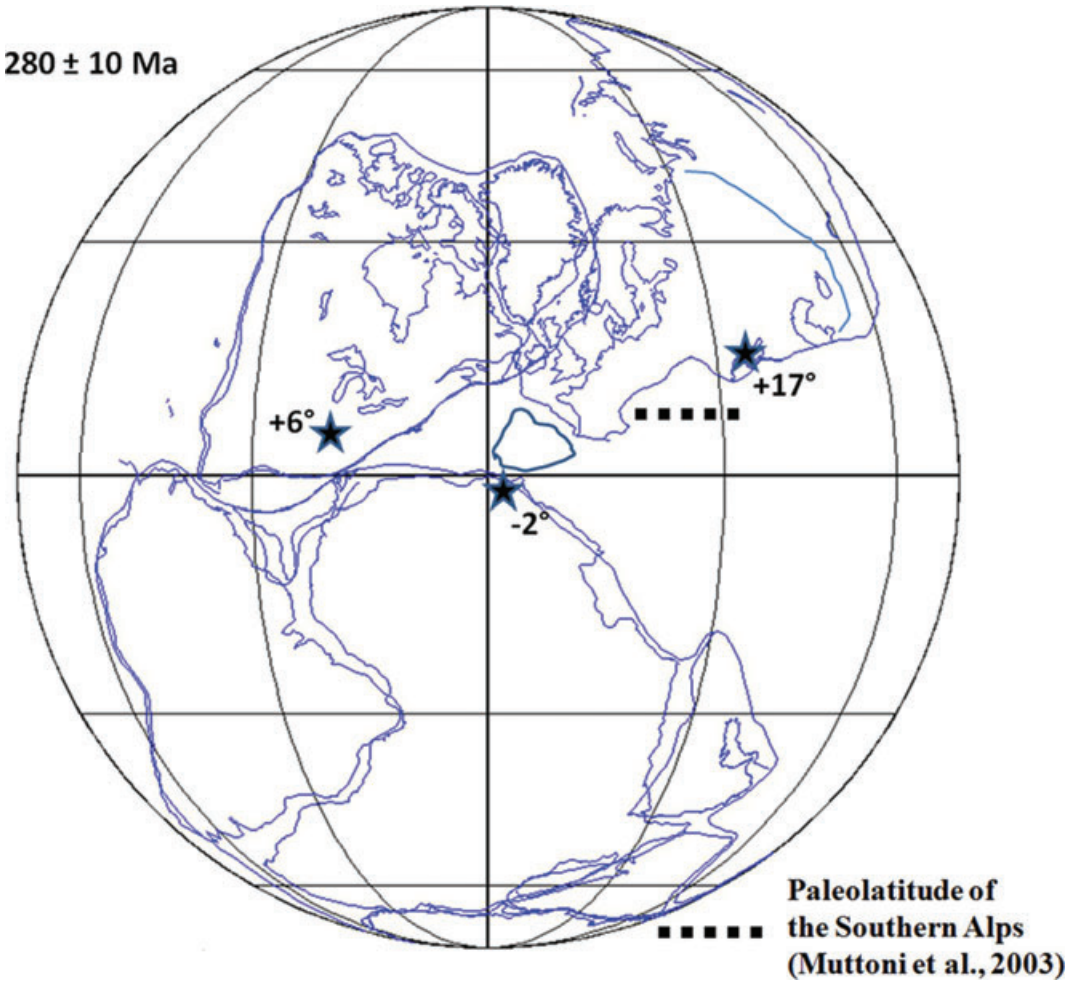


Figure 11. Palaeogeographical map showing Baltica, positioned according to its mean palaeopole for 280 ± 10 Ma (from Table 4, at 51.2°N, 169.3°E), Gondwana, positioned according to its running-mean pole for 280 Ma (from Table 4 of Torsvik & Van der Voo 2002), and Laurentia, based on the palaeopole from 268–273 Ma igneous rocks from Illinois (Reynolds *et al.* 1997). The stars represent the sampling localities in Ukraine, Morocco and Illinois, where igneous rocks with ages within 280 ± 10 Ma have yielded palaeomagnetic palaeolatitudes. The dashed line shows the palaeolatitude where the mean Permian pole for greater Gondwana of Muttoni *et al.* (2003) would position the Southern Alps of Italy.

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APPENDIX

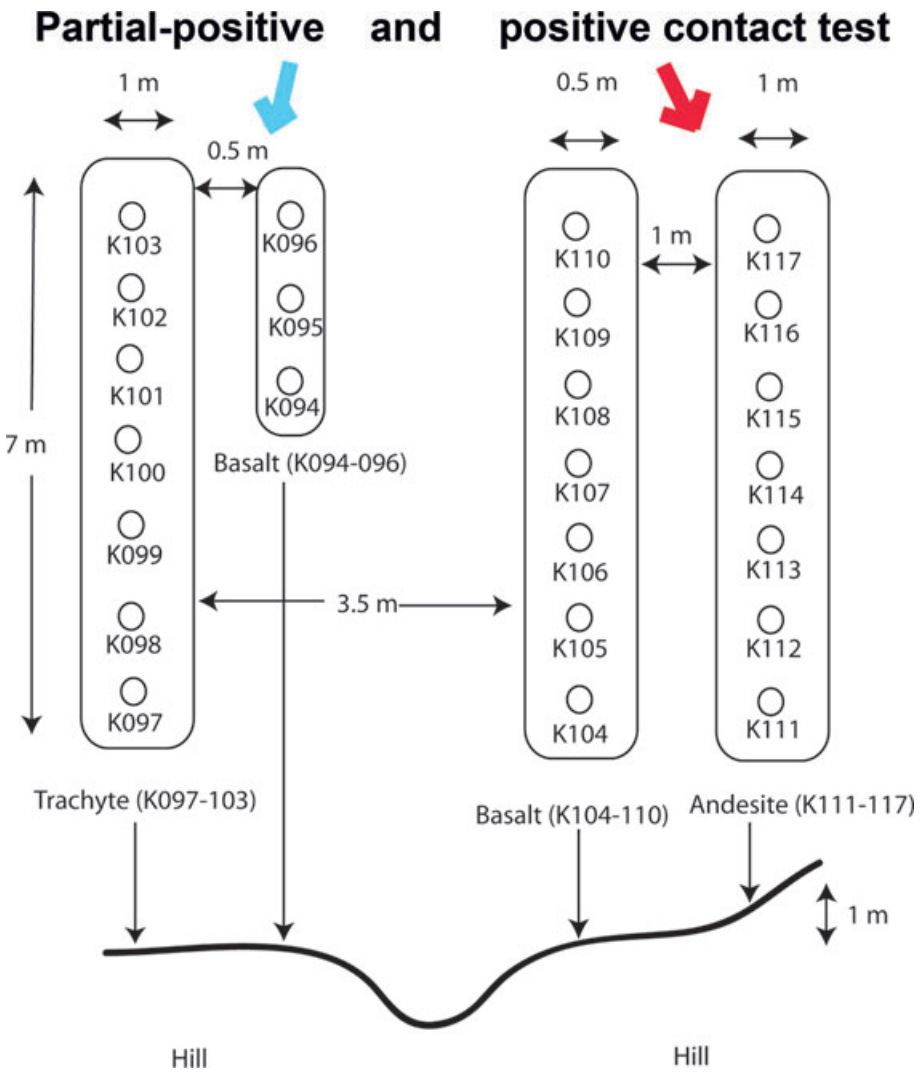


Figure A1. Very schematic map of the spatial relationships between a Late Triassic andesite intrusion, a mid-Early Permian trachyte dyke and basaltic host rocks of Devonian age, with sample numbers and envelopes for each of the sites.

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Table A1. Complete data tables for $^{40}Ar/^{39}Ar$ analyses. (1) ^{39}Ar (x 10^{-16} mols). (2) Cumulative ^{39}Ar gas released during the experiment (per cent). (3) %40* = per cent radiogenic ^{40}Ar of total ^{40}Ar released.

M1171	$J = 0.0058810$									
Temp	40/39	38/39	37/39	36/39	39Ar (1)	F39Ar (2)	%40* (3)	40*/39K	Age	± 1 s.d.
439	83.346	0.105	0.861	151.801	0.500	0.15	46.20	38.52	368.60	10.82
582	728.331	0.597	2.197	2378.149	1.089	0.49	3.50	25.71	254.12	37.19
662	123.208	0.190	2.559	327.092	0.740	0.72	21.70	26.70	263.24	7.76
762	122.675	0.207	3.204	303.001	1.007	1.03	27.20	33.34	323.21	8.38
876	35.683	0.179	2.604	16.451	5.145	2.62	86.80	30.98	302.09	1.10
987	29.345	0.195	2.432	1.767	177.787	57.55	98.70	28.96	283.90	1.34
1055	29.062	0.191	2.459	1.287	89.739	85.28	99.20	28.82	282.64	0.40
1067	30.210	0.193	3.130	8.852	3.980	86.51	92.00	27.78	273.15	1.72
1081	30.663	0.208	3.159	10.753	2.548	87.29	90.20	27.67	272.17	3.15
1119	29.988	0.182	2.873	6.295	5.363	88.95	94.40	28.30	277.86	1.52
1155	29.300	0.181	2.863	2.519	30.489	98.37	98.00	28.72	281.75	0.38
1225	31.254	0.173	2.981	9.310	3.048	99.31	91.80	28.68	281.34	2.25
1288	32.728	0.192	2.854	6.822	2.223	100.00	94.40	30.89	301.26	3.10
M1199	$J = 0.0058800$									
Temp	40/39	38/39	37/39	36/39	39Ar (1)	F39Ar (2)	%40* (3)	40*/39K	Age	± 1 s.d.
435	37.368	0.077	0.610	87.885	0.740	0.71	30.50	11.40	117.12	6.74
580	453.512	0.341	1.350	1477.318	1.805	2.43	3.80	17.02	172.13	15.01
661	39.318	0.030	0.475	78.252	1.207	3.58	41.20	16.19	164.15	2.66
763	180.390	0.163	0.675	550.480	1.061	4.59	9.80	17.73	179.03	8.61
876	43.962	0.132	2.353	88.087	0.661	5.22	41.10	18.05	182.05	10.67
986	48.612	0.783	13.804	98.123	1.341	6.50	42.10	20.47	205.18	5.32
1029	26.332	0.908	18.961	24.783	3.455	9.79	76.70	20.19	202.52	2.01
1053	23.963	0.927	20.176	16.480	19.238	28.12	85.00	20.36	204.09	0.46
1065	21.910	0.929	20.616	10.434	24.525	51.50	91.80	20.12	201.79	0.32
1079	21.706	0.917	20.955	8.460	31.577	81.60	94.60	20.52	205.66	0.25
1117	24.772	0.904	20.666	20.754	13.932	94.87	80.40	19.93	200.02	0.79
1153	28.905	1.002	20.492	35.417	1.216	96.03	68.20	19.71	197.98	4.79
1225	31.212	1.035	20.329	44.179	1.728	97.68	62.20	19.42	195.15	3.25
1288	29.213	1.045	21.578	38.574	2.435	100.00	65.60	19.15	192.60	2.47
M1208	$J = 0.0058340$									
Temp	40/39	38/39	37/39	36/39	39Ar (1)	F39Ar (2)	%40* (3)	40*/39K	Age	± 1 s.d.
413	116.815	0.300	1.108	349.337	0.179	0.32	11.70	13.62	138.00	29.03
548	98.820	0.128	0.821	248.456	1.140	2.33	25.70	25.42	249.65	7.02
615	1055.256	0.768	1.737	3469.393	0.814	3.76	2.90	30.14	292.40	35.90
694	51.573	0.101	0.729	96.002	0.601	4.82	45.00	23.22	229.32	9.42
765	80.108	0.163	1.436	161.433	0.294	5.34	40.50	32.48	313.19	18.31
880	186.169	0.690	6.128	587.631	0.306	5.88	6.90	12.85	130.51	24.58
984	47.493	0.910	15.452	92.807	2.567	10.41	44.30	21.04	208.96	3.98
1024	32.201	0.775	16.370	41.109	7.771	24.11	65.50	21.08	209.38	1.12
1050	26.842	0.730	15.824	23.505	13.238	47.45	77.80	20.89	207.55	0.67
1072	24.317	0.762	16.740	14.588	14.354	72.76	86.60	21.06	209.16	0.59
1102	22.598	0.737	16.916	8.767	12.165	94.21	93.20	21.07	209.28	0.69
1123	33.603	0.907	16.210	45.512	0.581	95.24	63.00	21.17	210.24	11.78
1154	33.894	0.888	17.107	37.965	0.528	96.17	70.20	23.79	234.56	10.25
1212	37.664	0.810	16.268	48.310	0.334	96.76	64.90	24.45	240.70	12.38
1289	27.574	0.775	15.820	30.049	1.840	100.00	71.30	19.67	196.10	1.77
M1213	$J = 0.0058780$									
Temp	40/39	38/39	37/39	36/39	39Ar (1)	F39Ar (2)	%40* (3)	40*/39K	Age	±1 s.d.
429	289.479	0.285	2.596	985.241	0.179	0.26	0.00	0.00	0.00	0.00
536	103.892	0.124	1.117	300.972	0.553	1.08	14.40	14.99	152.42	27.09
602	1460.017	1.120	2.709	4826.229	0.607	1.97	2.30	34.03	329.19	105.99
658	153.009	0.172	0.685	432.625	0.444	2.63	16.50	25.18	249.17	28.07
717	199.520	0.243	1.074	607.296	0.303	3.07	10.10	20.10	201.58	47.85
797	386.097	0.444	1.687	1168.541	0.213	3.39	10.60	40.89	388.83	178.62

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Table A1. *(Continued.)*

M1213	$J = 0.0058780$									
Temp	40/39	38/39	37/39	36/39	39Ar [(1)]	F39Ar [(2)]	%40* [(3)]	40*/39K	Age	±1 s.d.
863	148.486	0.486	3.517	446.643	0.112	3.55	11.20	16.69	168.90	80.86
934	124.968	1.043	11.126	381.670	0.165	3.80	10.20	12.81	131.00	42.66
1000	40.569	0.922	15.068	82.018	0.960	5.21	42.50	17.23	174.17	4.83
1085	22.654	0.731	16.431	9.765	49.400	78.03	91.80	20.80	208.19	0.39
1153	22.519	0.725	16.699	9.335	13.993	98.66	92.40	20.81	208.28	0.76
1198	67.076	1.043	16.838	168.448	0.404	99.26	27.30	18.32	184.65	22.04
1263	88.578	1.046	17.691	242.909	0.503	100.00	20.20	17.87	180.30	18.71
M1215	$J = 0.0058600$									
Temp	40/39	38/39	37/39	36/39	39Ar [(1)]	F39Ar [(2)]	%40* [(3)]	40*/39K	Age	± 1 s.d.
426	90.386	0.387	0.772	247.686	0.267	0.42	19.00	17.21	173.43	26.70
555	70.280	0.113	0.865	171.161	1.149	2.23	28.10	19.72	197.44	4.17
616	715.599	0.494	1.558	2362.266	0.847	3.56	2.50	17.61	177.31	61.39
675	58.924	0.102	0.863	126.118	0.516	4.38	36.80	21.68	215.88	8.75
744	57.983	0.174	1.593	155.486	0.327	4.89	20.90	12.10	123.63	13.84
826	104.858	0.363	2.860	313.998	0.260	5.30	11.60	12.21	124.70	26.76
921	131.333	0.864	10.603	357.924	0.387	5.91	20.00	26.26	258.37	17.29
987	42.841	0.858	15.162	82.883	1.883	8.87	45.00	19.28	193.21	3.56
1032	24.201	0.715	16.452	14.767	11.729	27.34	86.20	20.87	208.25	0.74
1075	23.252	0.719	16.644	10.808	30.193	74.89	90.80	21.10	210.47	0.43
1111	24.813	0.705	16.995	17.027	14.482	97.70	84.00	20.85	208.06	0.81
1153	96.735	0.996	17.752	264.008	0.588	98.62	20.50	19.82	198.36	17.11
1210	186.046	0.920	17.390	593.498	0.291	99.08	6.30	11.64	119.12	29.33
1289	163.622	0.898	17.153	462.511	0.585	100.00	17.20	28.12	275.32	13.98
M1226	$J = 0.0058840$									
Temp	40/39	38/39	37/39	36/39	39Ar [(1)]	F39Ar [(2)]	%40* [(3)]	40*/39K	Age	± 1 s.d.
430	259.703	0.207	1.361	775.614	0.117	0.09	11.80	30.57	298.61	70.01
558	565.019	0.556	2.513	1812.691	0.521	0.51	5.20	29.51	289.04	68.38
620	909.284	0.677	1.817	2935.943	0.272	0.73	4.60	41.83	397.14	122.79
695	1463.207	1.064	1.156	4813.171	0.260	0.93	2.80	40.97	389.88	137.67
778	172.483	0.233	0.991	545.605	0.341	1.21	6.50	11.28	116.00	24.14
881	81.614	0.652	3.938	222.026	0.282	1.43	19.90	16.21	164.48	21.14
985	35.468	1.154	12.189	51.456	2.520	3.44	59.30	21.02	210.48	1.97
1028	23.938	1.152	13.190	12.364	9.220	10.79	88.20	21.11	211.31	0.51
1052	22.171	1.147	13.414	7.069	32.666	36.85	94.30	20.92	209.52	0.52
1079	22.323	1.093	13.428	5.648	55.356	80.99	96.30	21.50	214.98	0.17
1116	22.686	0.968	13.566	6.945	13.596	91.84	94.70	21.49	214.88	0.38
1153	26.524	1.092	13.258	19.892	1.254	92.84	81.00	21.48	214.80	3.35
1211	24.354	1.255	13.873	13.433	2.950	95.19	87.30	21.25	212.69	1.45
1289	24.343	1.202	13.863	14.850	6.029	100.00	85.50	20.82	208.58	0.55
M1234	$J = 0.0058570$									
Temp	40/39	38/39	37/39	36/39	39Ar [(1)]	F39Ar [(2)]	%40* [(3)]	40*/39K	Age	± 1 s.d.
428	187.722	0.151	1.462	568.867	0.369	0.43	10.50	19.68	196.95	17.24
556	98.771	0.121	1.221	264.248	0.989	1.57	21.00	20.73	206.86	7.26
617	838.751	0.609	1.599	2717.563	0.495	2.14	4.30	35.80	343.60	62.40
677	93.389	0.098	0.880	225.784	0.286	2.47	28.60	26.70	262.25	19.29
744	82.031	0.132	1.288	222.653	0.176	2.67	19.90	16.28	164.43	32.28
812	120.293	0.409	2.563	355.145	0.128	2.82	12.90	15.47	156.57	35.03
920	109.438	0.848	10.169	309.228	0.433	3.32	17.10	18.67	187.31	18.30
986	37.263	0.617	13.679	58.387	3.201	7.02	56.00	20.86	208.09	1.20
1030	23.563	0.502	13.996	11.966	21.752	32.15	88.70	20.90	208.45	0.32
1073	28.549	0.537	14.417	28.617	43.169	82.03	73.50	20.99	209.33	0.29
1111	25.429	0.561	15.177	19.014	12.685	96.68	81.60	20.76	207.11	0.38
1154	28.112	0.704	14.207	45.126	0.911	97.73	55.50	15.61	157.90	4.16
1212	36.010	0.553	11.702	92.079	0.531	98.35	26.20	9.43	97.00	11.88
1264	39.192	0.659	13.829	75.656	1.429	100.00	45.10	17.67	177.72	4.29

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